

11QL 2º bndado

$$2.1 \quad |+\rangle = 1/\sqrt{2} (|0\rangle + |1\rangle) = 1/\sqrt{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$|-\rangle = 1/\sqrt{2} (|0\rangle - |1\rangle) = 1/\sqrt{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$a) \quad |0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{bmatrix} 1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{bmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$b) \quad |0\rangle \otimes |+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes 1/\sqrt{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1/\sqrt{2} \begin{bmatrix} 1 \times \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ 0 \times \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{bmatrix} = 1/\sqrt{2} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$c) \quad |+\rangle \otimes |1\rangle = 1/\sqrt{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1/\sqrt{2} \begin{bmatrix} 1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{bmatrix} = 1/\sqrt{2} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

$$d) \quad |-\rangle \otimes |+\rangle = 1/\sqrt{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \otimes 1/\sqrt{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1/2 \begin{bmatrix} 1 \times \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ -1 \times \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{bmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

$$2) \quad |\psi\rangle = 1/\sqrt{2} |00\rangle + 1/\sqrt{2} |01\rangle \text{ como 2 qubits separados}$$

$$|\psi\rangle = |0\rangle \otimes (1/\sqrt{2} |0\rangle + i/\sqrt{2} |1\rangle)$$

2.2

1)

matrizes (U) de um circuito de um qubit criadas pelas portas $U = XZH$

$$H = 1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$X \times Z \times H = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times 1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \times 1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = 1/\sqrt{2} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

2)

$$U = X \otimes Z \otimes H = \begin{bmatrix} 0 \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} & 1 \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ 1 \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} & 0 \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \otimes 1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$